De La Salle Holy Cross College

NOVEMBER EXAMINATION

2024

Grade 10

PHYSICAL SCIENCES

Paper 1: Physics

Examiner:	Mrs A. Kamedza	**Time:**	2 hours
Moderator:	Mr E. Hove	**Marks:**	130

Name: _____

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of 20 pages as well as a Data Sheet of 1 page. Please check that your question paper is complete.
2. Answer ALL the questions on the question paper.
3. Where appropriate take your answers to two decimal places.
4. Use data and formulae whenever necessary.
5. Show your working in all calculations.
6. It is in your interest to write legibly and to present your work neatly.
7. Pages 17 and 20 have been left blank. If you run out of space for an answer, use these pages. Clearly indicate the number of your answer should you use this space.

QUESTION NUMBER	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**TOTAL**
ACTUAL MARK									
TOTAL MARK	**20**	**12**	**13**	**19**	**20**	**9**	**22**	**15**	**130**

Remember that God is with you and will help and inspire you in all that you do

QUESTION 1 MULTIPLE CHOICE

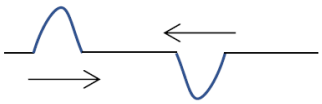
Answer the multiple-choice questions on the answer grid below. Make a clear cross (X) in the box corresponding to the letter that you consider to be correct. Every question has only one correct answer.

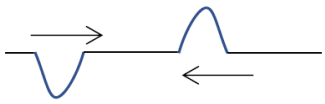
A	B	C	D

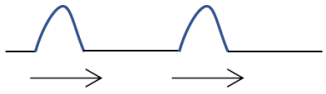
Here the option C has been marked as an example.

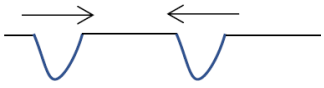
1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

1.1 Which of the following diagrams best illustrates constructive interference (pulses are travelling at the same speed)?

A 

B 

C 

D 

1.2 A photon of a microwave possesses 4,88 x 10^{-23} J of energy. It will then have the following frequency and wavelength:

	Frequency (Hz)	Wavelength (m)
A	3,0 x 10^{8}	6,6 x 10^{-34}
B	1,35 x 10^{-11}	2,22 x 10^{19}
C	7,39 x 10^{10}	4,06 x 10^{-3}
D	4,06 x 10^{-3}	7,39 x 10^{10}

1.3 A magnet is broken into two pieces as shown below:



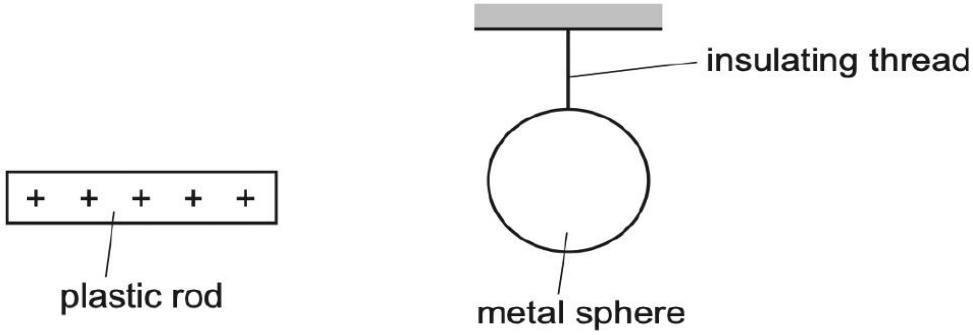
Which one of the following statements is true about the force between the broken pieces if they face each other?

A There is an attractive force between the broken pieces.
B There is a repulsive force between the broken pieces.
C There is an attractive force and then a repulsive force between the broken pieces.
D There is no force between the broken pieces since they are demagnetised.

1.4 A steel piston carries a charge of - 8 x 10^{-17} C. How many excess electrons does the piston have?

A 0
B 0,002
C 500
D 2500

1.5 A charged plastic rod is brought near an uncharged metal sphere. The metal sphere is suspended on an insulating thread.



Which one of the following options is correct?

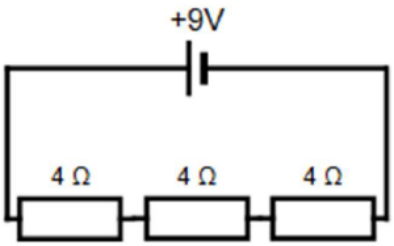
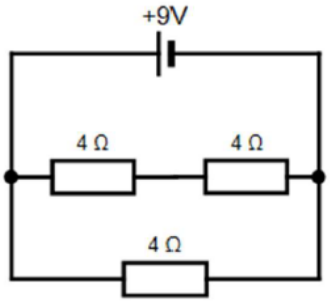
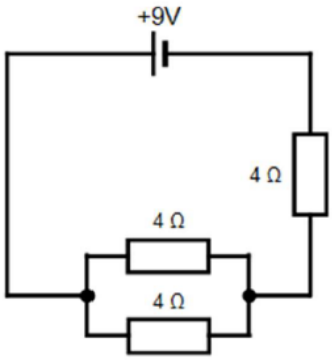
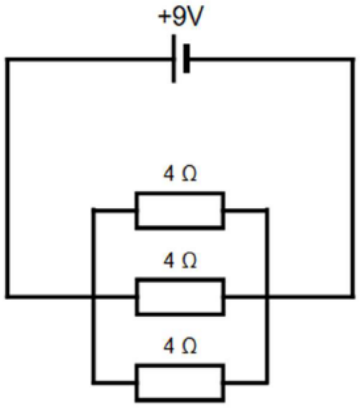
	Observation	State of the metal sphere
A	The rod attracts the sphere	charged
B	The rod attracts the sphere	uncharged
C	The rod repels the sphere	charged
D	The rod repels the sphere	uncharged

1.6 Which value of resistance **cannot** be made by combining three 10 Ω resistors?

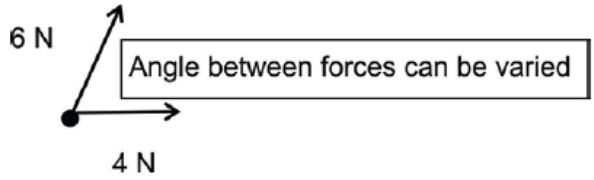
A 3,33 Ω
B 6,67 Ω
C 15,0 Ω
D 25,0 Ω

1.7 You have three identical resistors of 4 Ω each and a 9 V cell.

Which option illustrates the best way to connect these components so that the total current in the circuit is 1,5 A?

A	
B	
C	
D	

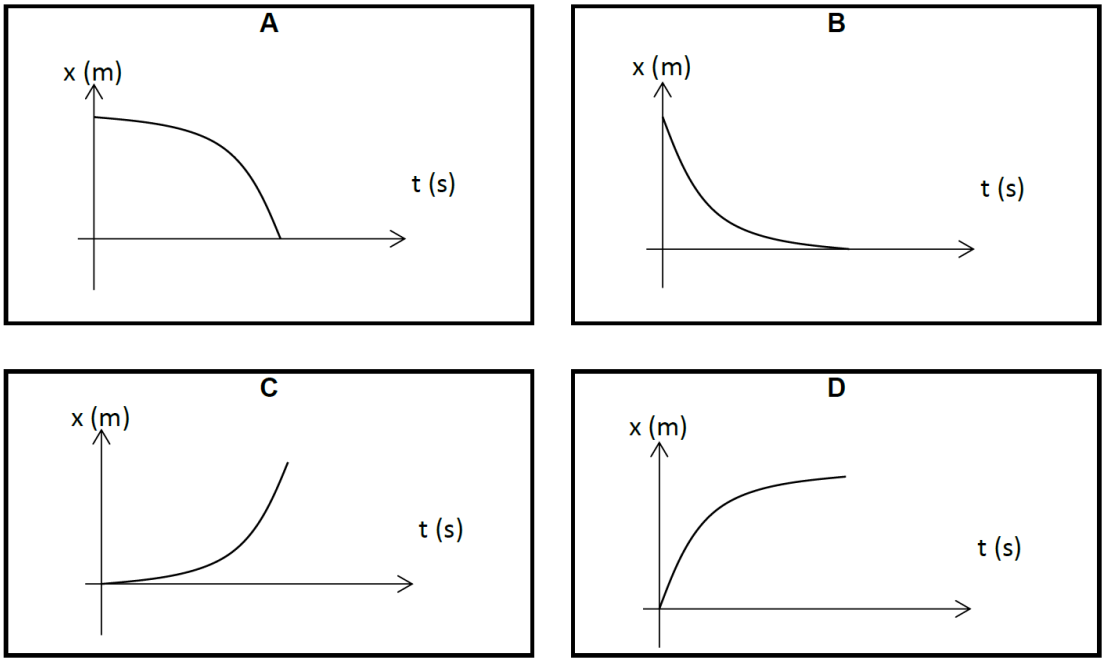
1.8 Two forces, one of 6 N and another of 4 N, act on the same object as shown in the diagram below. The angle between the forces can be varied/changed.



Which one of the following would **not** be able to bring these two forces into equilibrium, no matter the angle between the forces?

A 10 N
B 12 N
C 7 N
D 2 N

1.9 Which one of the following graphs best represents deceleration of a car moving backwards?



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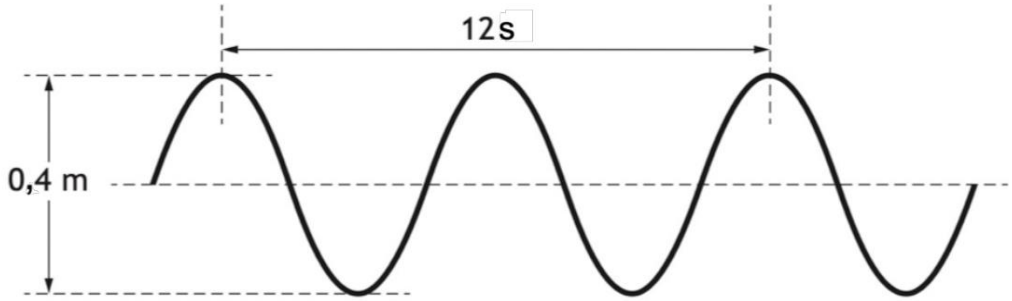
1.10 A truck has a kinetic energy of E_k. How is its kinetic energy affected if the truck doubles its velocity while halving its mass when it loses its load?

A $2\ E_k$
B E_k
C $\frac{1}{2}\ E_k$
D $4\ E_k$

[20]

QUESTION 2

2.1 The diagram below represents a water wave. The speed of the wave is 3,0 $m \cdot s^{-1}$.



2.1.1 What type of wave is a water wave? (1)

_____

2.1.2 Define *frequency*. (2)

_____

_____

_____

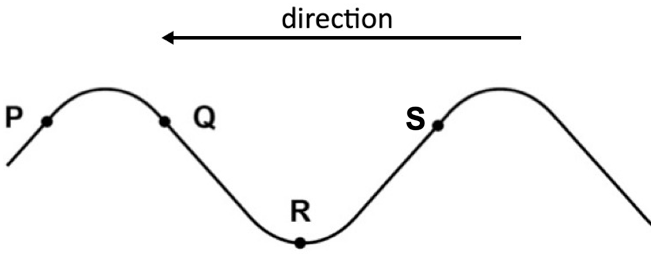
2.1.3 Write down the amplitude and frequency for this wave: (2)

Amplitude:

Frequency:

2.1.4 Hence, calculate the wavelength of this wave. (3)

2.2 The diagram below shows a wave in a string at one instant in time.



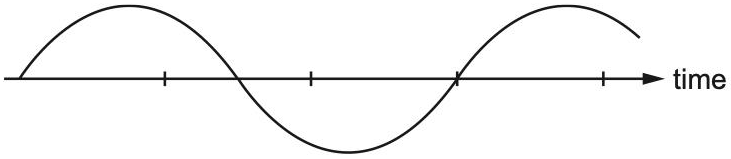
2.2.1 Identify two points on the wave that are in phase. (1)

_____

2.2.2 Identify two points on the wave that are out of phase. (1)

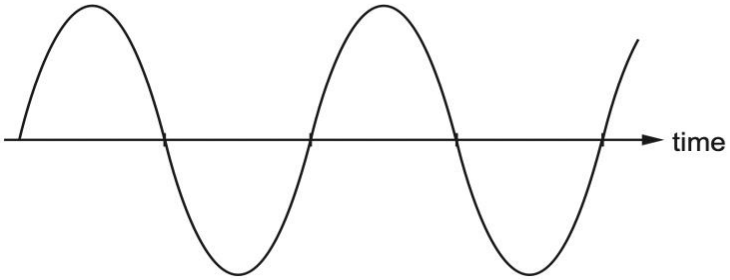
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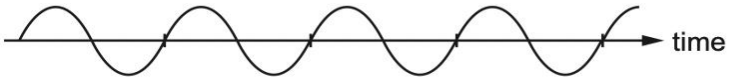
2.3 Consider the following diagram of a sound wave.

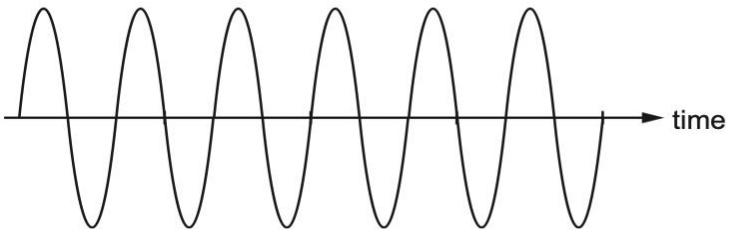


The diagrams below represent the waves produced by three other sources of sound. The scales are the same for all the diagrams.
Which wave represents a sound of higher pitch but lower volume than the original wave shown above?

Mark your choice by placing an X in the box. (2)

☐ 

☐ 

☐ 

[12]

QUESTION 3

3.1 What is meant by *electromagnetic radiation*? (3)

_____

_____

_____

_____

_____

3.2 The diagram below shows regions of the electromagnetic (EM) spectrum.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays

Identify **three** correct statements by placing an X in the corresponding boxes. (3)

All EM waves travel at the same speed in a vacuum.	☐
The energy of EM waves decreases from radio waves to gamma waves.	☐
Microwaves, ultraviolet and X-rays are all ionizing radiation.	☐
The wavelength of EM waves decreases from radio waves to gamma rays.	☐
Gamma rays can be emitted from radioactive materials.	☐
All EM waves are mechanical and longitudinal.	☐

3.3 Visible light with a wavelength of 7,8 nm is shone onto the surface of zinc and electrons are ejected.

3.3.1 What evidence does the observation above provide? (1)

_____

3.3.2 What are the particles of light called and what do we know about their energy? (2)

_____

_____

_____

3.3.3 Calculate the energy of this light. (4)

[13]

QUESTION 4

4.1 MRI machines employ powerful magnets which produce a strong magnetic field that forces protons in the body to align with the field.

4.1.1 What is meant by the term *magnetic field*? (2)

_____

_____

_____

_____

4.1.2 Explain how an electric current can turn an iron nail into a magnet. (3)



[Image source: https://carnegiestemgirls.org/wp-content/uploads/2016/04/nail-electromagnet-300x254.jpg]

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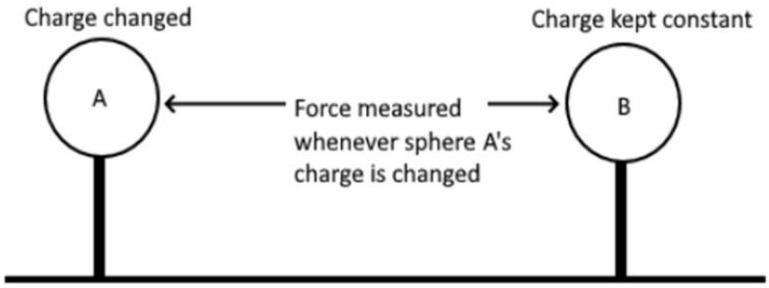
_____

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_____

_____

4.2 Two identical spheres are used in an experiment to determine the relationship between the force between them and the magnitude of the charge.



https://images.app.goo.gl/QMWR5pk1W6oEY9pa6 Accessed 20/10/2024

The magnitude of the charge on sphere A is changed while the magnitude of the charge on sphere B is kept constant. The distance between the spheres is also kept constant.

The results of the experiment are given below.

Charge on Sphere A (C)	Force on Sphere B (N)
0,00	0
0,15	2
0,32	4
0,45	6
0,57	8
0,72	10

4.2.1 State the variables below: (4)

Independent Variable: _____

Dependent variable: _____

2 Controlled variables: _____

_____

4.2.2 Write down the hypothesis for the investigation. (2)

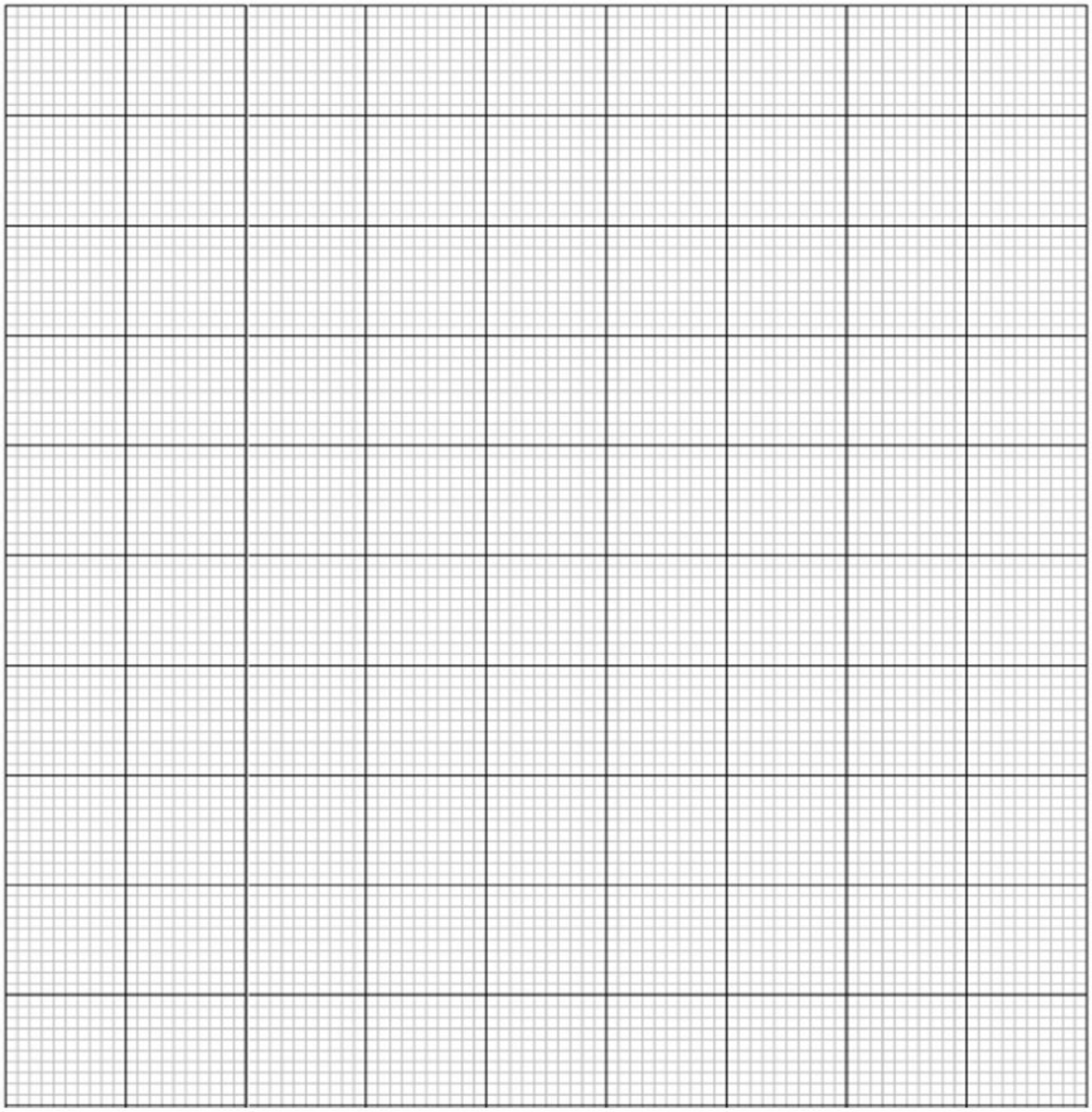
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_____

4.2.3 Plot a graph of the force on sphere B versus the charge on sphere A. (6)



4.2.4 State the relationship between the force and the charge on the sphere. (2)

_____

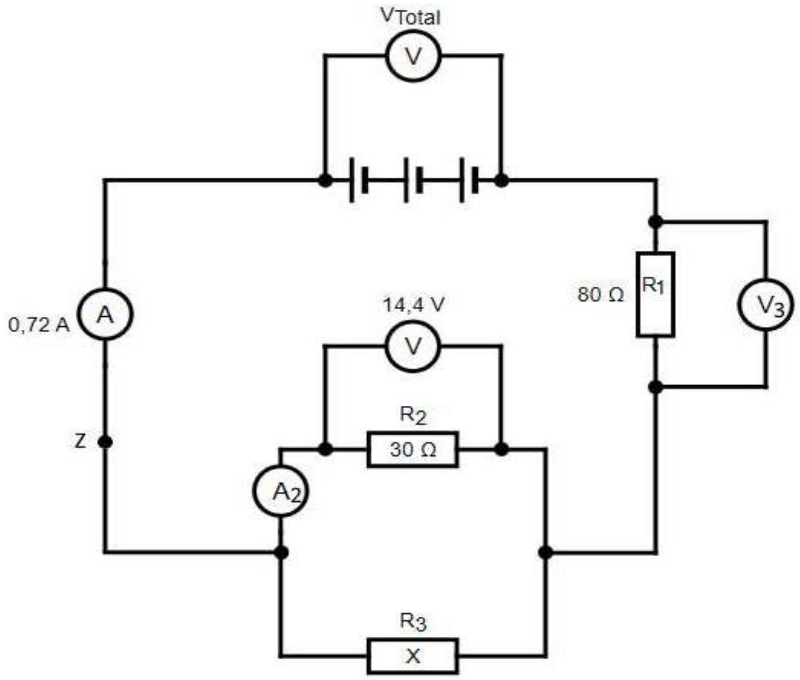
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_____

[19]

QUESTION 5

Use the circuit diagram below to answer the questions that follow:



5.1 Define the following terms:

5.1.1 Define *current strength*. (2)

_____

_____

5.1.2 State *Ohm's law*. (2)

_____

_____

_____

_____

5.2 Calculate the current reading on ammeter A_2. (3)

5.3 Determine the resistance of resistor X. (5)

5.4 State the reading on voltmeter V_3. (2)

5.5 Determine the total voltage (V_{TOTAL}) across the battery. (2)

5.6 Calculate the charge that flows past point Z in 2,5 minutes. (4)

[20]

QUESTION 6

A wooden block is pushed with a force of 30 N North and 20 N East.

6.1 Define a *vector quantity*. (2)

_____

_____

6.2 Use either a vector diagram drawn to scale or an algebraic method to determine the resultant force. (5)

6.3 Thus, write down the equilibrant force using your answer to Question 6.2. (2)

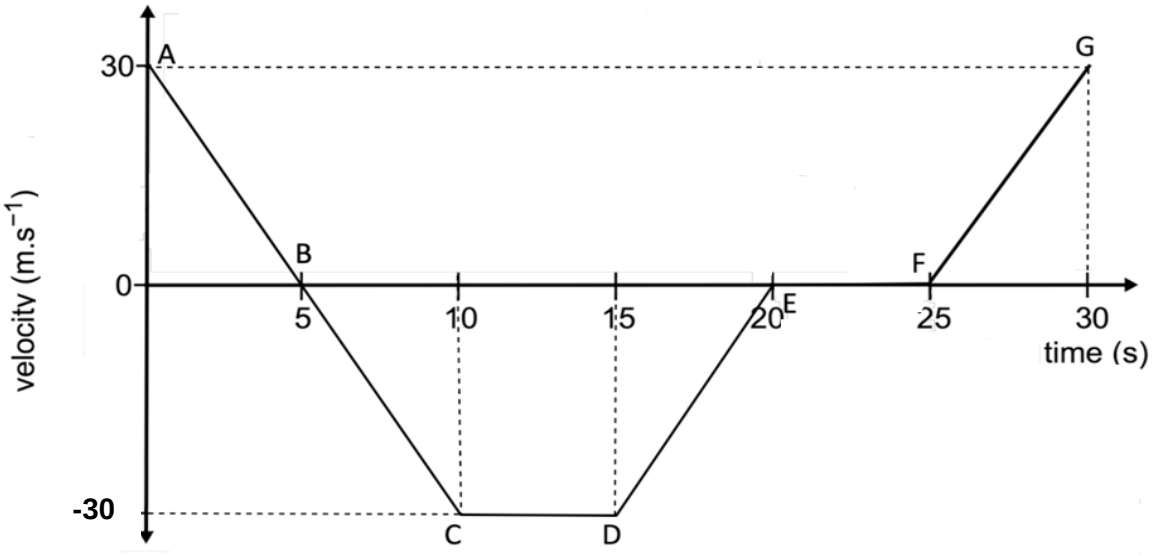
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[9]

QUESTION 7

A car is initially travelling **east** along a straight road. A velocity vs. time graph of its motion is given below.



7.1 Define *displacement*. (1)

_____

_____

7.2 At point **G**, after 30 s, is the car's displacement east or west?
Explain your answer. (3)

_____

_____

_____

7.3 Calculate the car's average velocity for the first 15 seconds. (6)

7.4 During which interval(s) (A - G) is the car:

7.4.1 travelling east? (1)

_____

7.4.2 experiencing constant negative acceleration? (1)

_____

7.4.3 stationary for more than one second? (1)

_____

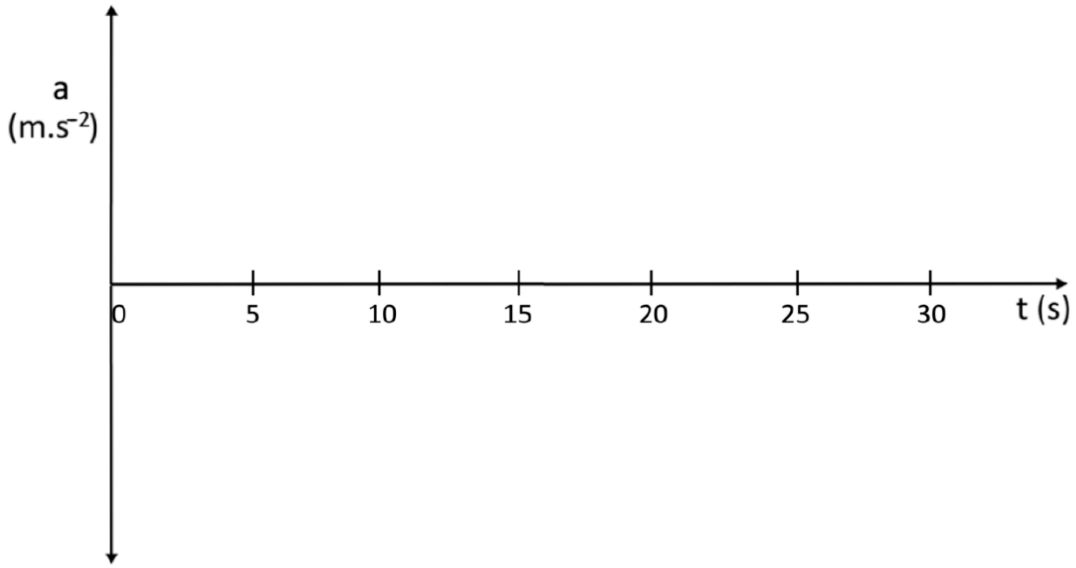
7.5 Define *acceleration*. (2)

_____

_____

7.6 Calculate the car's acceleration between 15 s and 20 s. (4)

7.7 Hence, draw an acceleration vs. time graph for the 30 second period. (3)
No values are needed but the graph must correlate to the time values on the x-axis.



[22]

ADDITIONAL SPACE (All questions)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.

QUESTION 8

Our entire universe is governed by energy. It manifests in many different forms, but its origins remain a mystery.

8.1 Define *gravitational potential energy*. (2)

_____

_____

_____

_____

8.2 Consider the following two situations:

1. A 270 kg gorilla runs at you at 5 $m \cdot s^{-1}$.
2. A 25 kg badger runs at you at 25 $m \cdot s^{-1}$.

Which situation indicates the animal with the most kinetic energy?
Use calculations to explain your answer. (5)

8.3 State the *principle of conservation of mechanical energy*. (2)

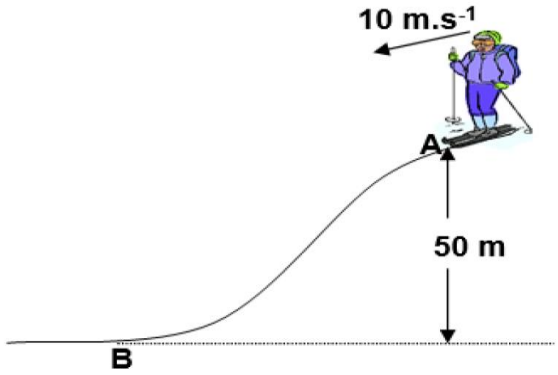
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8.4 A skier, including her equipment, with a total mass of 80 kg, approaches the edge of a ski slope 50 m high at a speed of 10 $m \cdot s^{-1}$ as shown in the diagram below.
Ignore air and surface friction.



https://www.dccreationsresources.co.za/subjects/physical-sciences/grade-10-physical-sciences/lessons/mid-year-assessments-18/

Calculate the speed of the skier at the bottom of the slope. (6)

[15]

TOTAL: 130 marks

ADDITIONAL SPACE (All questions)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.